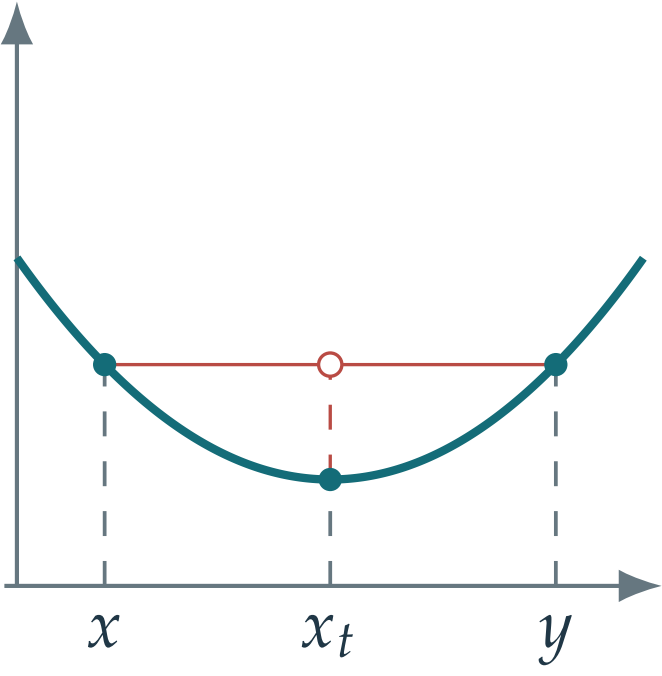
Convex



$$f(x_t) \leq \ell_t$$